

SUMIF Function in Excel

1. What is SUMIF?

The SUMIF function adds only those numbers that meet one condition. It solves the problem of finding a total for one group inside a larger list, such as total sales for one region or total cost for one expense category. A Data Analyst uses it to build group totals without filtering or copying the data.

2. Syntax

`=SUMIF(range, criteria, [sum_range])`

Argument	Required?	What it accepts
range	Required	The cells Excel checks against the condition (for example B2:B10)
criteria	Required	The condition to match: text ("North"), a number (100), a comparison (">=500"), or a cell reference
sum_range	Optional	The cells that are actually added (for example C2:C10). If left out, Excel adds the cells in range itself.

Use the third argument when the condition is in one column and the numbers to add are in another. If both are the same column, you can leave it out: `=SUMIF(B2:B7, ">=500")`.

3. Simple Example

Find the total sales for the North region:

	A	B	C
1	Product Category	Region	Sales
2	Laptop	North	5000
3	Laptop	South	7000
4	Laptop	East	6000
5	Monitor	North	3000
6	Monitor	South	4500
7	Monitor	East	3800
8	Mouse	North	2500
9	Mouse	South	3200
10	Mouse	East	2800
11	Total	North	10500

Formula in C11: `=SUMIF(B2:B10, "North", C2:C10)`

Excel looks for "North" in B2:B10 (rows 2, 5, 8), takes the matching sales from C2:C10 (5000, 3000, 2500), and adds them. **Result: 10500**

4. Another Example

Find the total of expenses that are 500 or more:

	A	B
1	Expense	Amount
2	Rent	1200
3	Internet	300
4	Travel	650
5	Supplies	150
6	Software	500
7	Snacks	90
8	Total (>=500)	2350

Formula in B8: `=SUMIF(B2:B7, ">=500")`

Here the condition and the numbers to add are in the same column, so no third argument is needed. Excel finds the amounts of 500 or more (1200, 650, 500) and adds them. **Result: 2350**

5. When to Use It

- Calculate total sales for one region or product
- Calculate total expenses for one category
- Calculate the total value of orders with a given status
- Calculate total units in stock for one product
- Calculate total hours or performance points for one team

6. Important Things to Remember

- SUMIF works with **one** condition only. For two or more conditions, use SUMIFS.
- Text and comparison criteria go inside quotation marks: "North", ">=500", "<>East". Matching is not case-sensitive.
- range and sum_range must be the same size, because Excel matches them row by row.
- Text and blank cells in sum_range are ignored.
- If no cell meets the condition, SUMIF returns 0. This can hide a spelling problem, so check unexpected zeros.
- Wildcards work with text: "Lap*" matches any text starting with Lap. To use a cell value in a comparison, join it: ">="&E1.

7. Common Mistakes

Mistake: Forgetting quotation marks around text or comparison criteria, such as North or >=500. **Fix:** Write "North" and ">=500".

Mistake: Using ranges of different sizes, such as B2:B10 with C2:C9. **Fix:** Select the same rows in both ranges.

Mistake: Swapping the order and putting the numbers first. **Fix:** Remember the order: condition range, condition, then numbers to add.

Mistake: Getting 0 because the text does not match exactly, such as "North " with an extra space. **Fix:** Clean the data first, or check the unique values in the column.

8. Data Analyst Tip

Type each group name in a cell and refer to it, for example `=SUMIF($B$2:$B$10, E2, $C$2:$C$10)`, then copy the formula down to build a summary by region. Always check your work: the group totals should add up to `=SUM(C2:C10)`. Here 10500 + 14700 + 12600 = 37800. If they do not match, a region name is misspelled or missing.